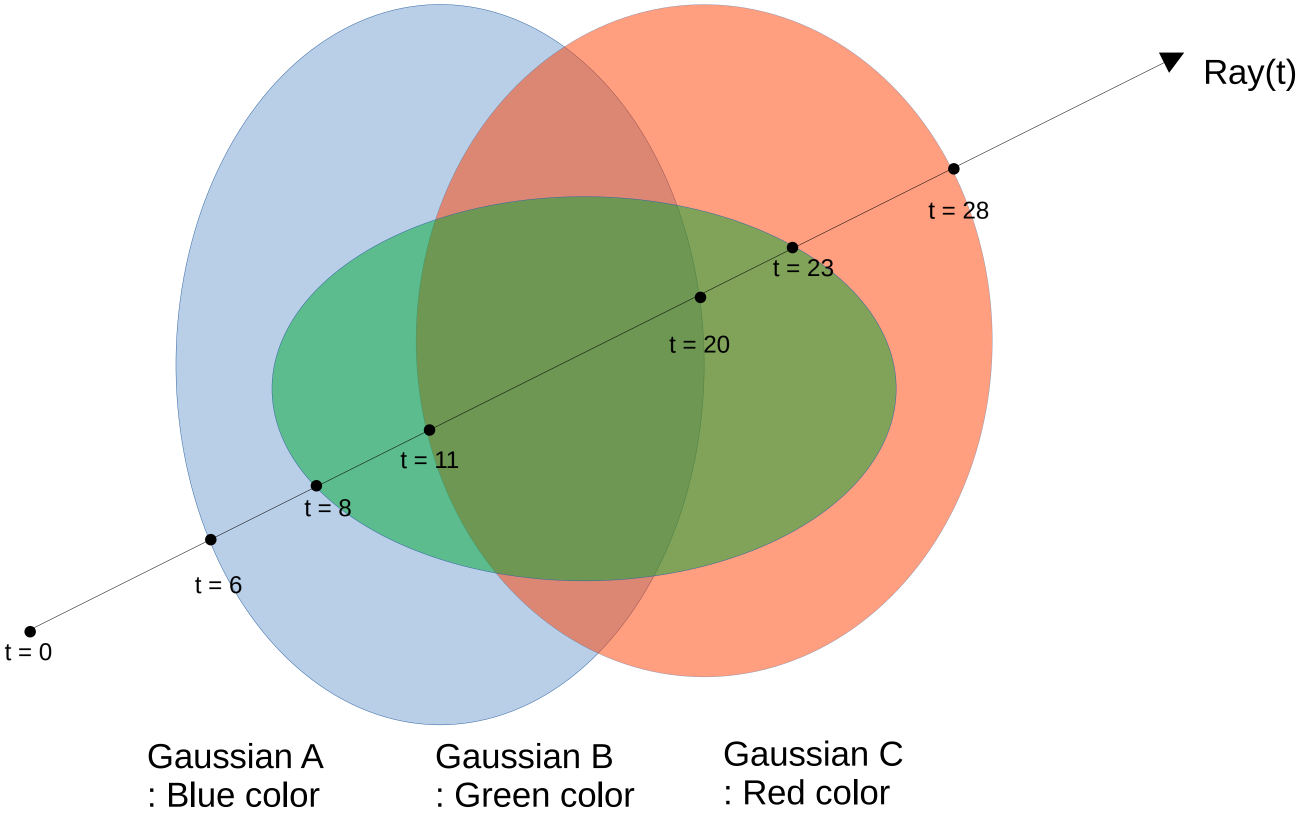
EVER Overlap Gaussian



Gaussian n 개일 때 event list 는 2n 개

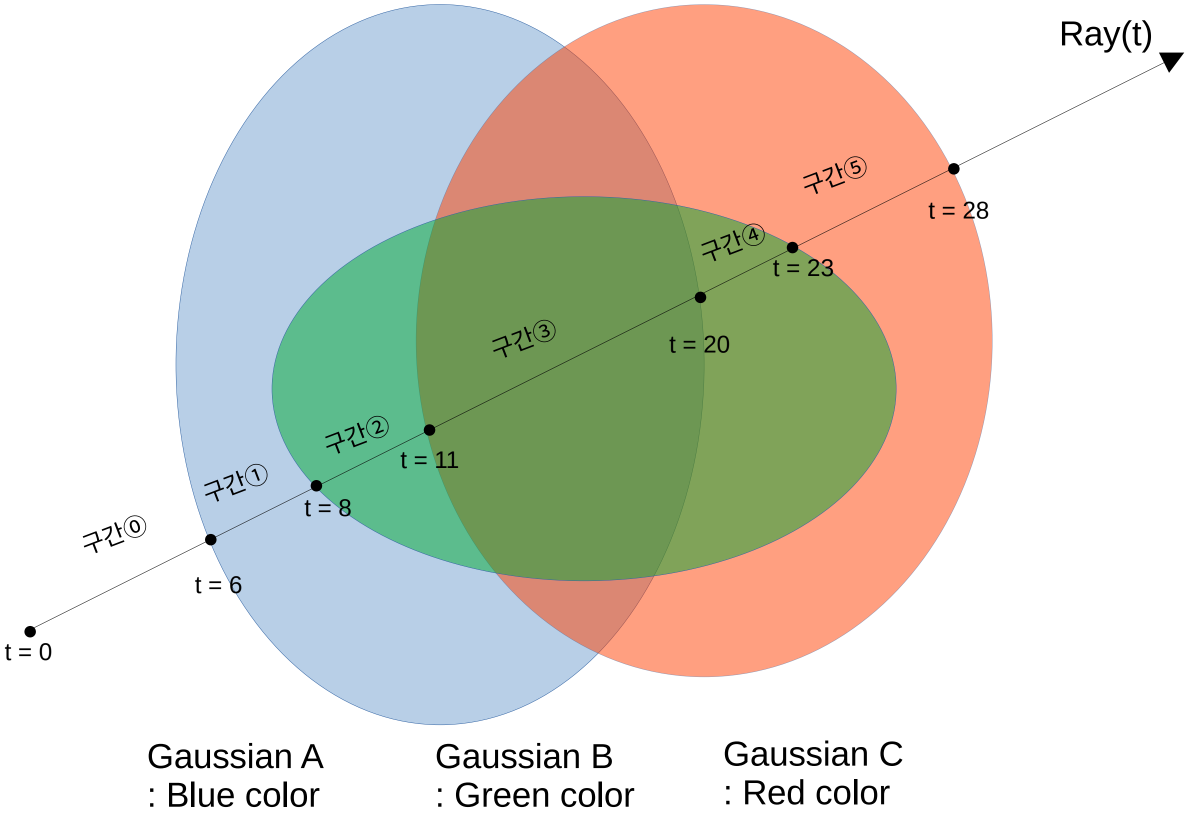
Gaussian 은 3 개

각 가우시안의 σ(밀도) 는 0.1 로 가정

Event list :

{A=6, B=8, C=11, -A=20, -B=23, -C=28}

EVER Overlap Gaussian



α = 1 - exp(-σ · Δt) : 구간 (t) 알파

c_blend = (σ_A·c_A + σ_B·c_B + ...) / σ_total : 구간 색

c_hat += T · α · c_blend : ray 전 구간 누적 색

각 가우시안의 σ(밀도) 는 0.1 로 가정

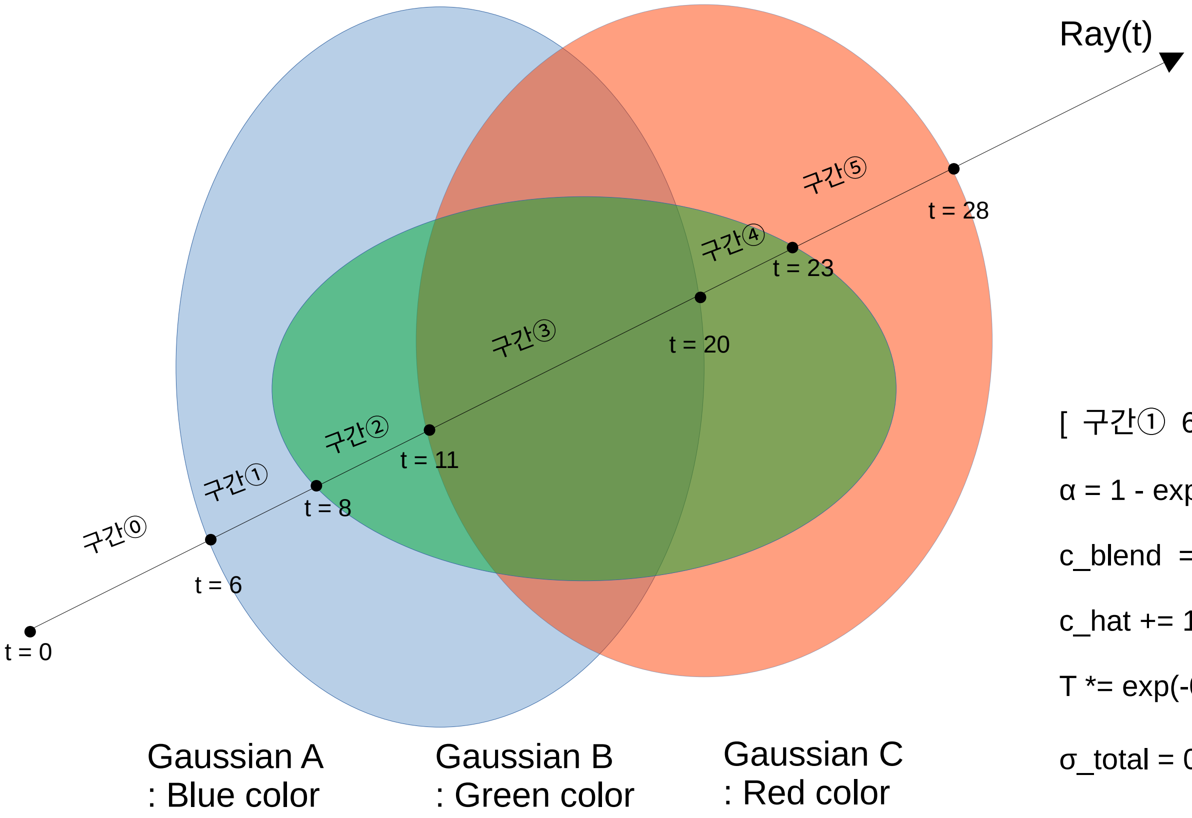
Event list :

{A=6, B=8, C=11, -A=20, -B=23, -C=28}

[구간① 0 → 6] active={}

초기값 : T = 1.000 / σ_total = 0 / c_hat = 0

EVER Overlap Gaussian



$\alpha = 1 - \exp(-\sigma \cdot \Delta t)$: 구간 (t) 알파

$c\_blend = (\sigma\_A \cdot c\_A + \sigma\_B \cdot c\_B + \ldots) / \sigma\_total$: 구간 색

$c\_hat \mathrel{+}= T \cdot \alpha \cdot c\_blend$: ray 전구간 누적 색

Event list : {A=6, B=8, C=11, -A=20, -B=23, -C=28}

각 가우시안의 σ(밀도) 는 0. 1 로 가정

[구간① 6 → 8] active={A} $\Delta t = 8-6 = 2$

$\alpha = 1 - \exp(-0.1 \times 2) = 1 - 0.819 = 0.181$

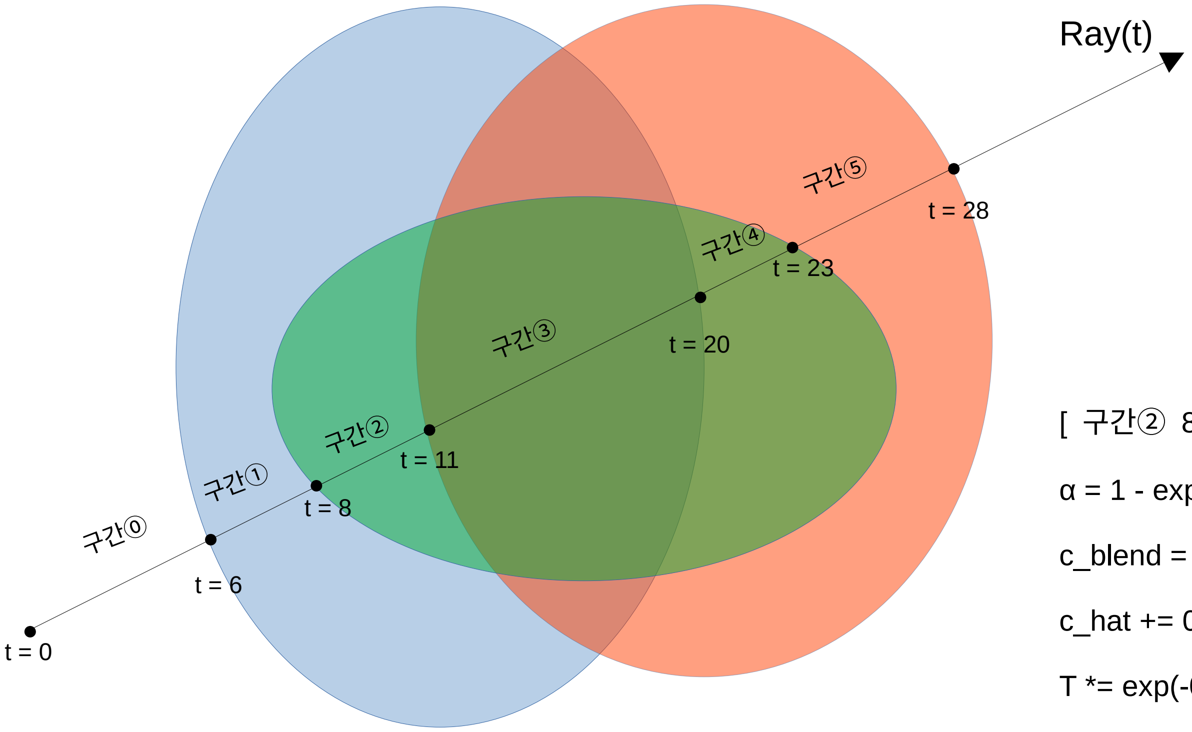
c_blend = Blue

c_hat += 1.000 × 0.181 × Blue

$T \mathrel{*}= \exp(-0.1 \times 2) = 1.000 \times 0.819 \rightarrow 0.819$

σ_total = 0 + 0.1 = 0.1 --> + B 진입으로 0.1 + 0.1 = 0.2

EVER Overlap Gaussian



Gaussian A
: Blue color

Gaussian B
: Green color

Gaussian C
: Red color

$\alpha = 1 - \exp(-\sigma \cdot \Delta t)$: 구간 (t) 알파

$c\_blend = (\sigma\_A \cdot c\_A + \sigma\_B \cdot c\_B + ...) / \sigma\_total$: 구간 색

$c\_hat \mathrel{+}= T \cdot \alpha \cdot c\_blend$: ray 전구간 누적 색

Event list : {A=6, B=8, C=11, -A=20, -B=23, -C=28}

각 가우시안의 σ(밀도) 는 0. 1 로 가정

[구간② 8 → 11] active={A, B} $\Delta t = 11-8 = 3$

$\alpha = 1 - \exp(-0.2 \times 3) = 1 - 0.549 = 0.451$

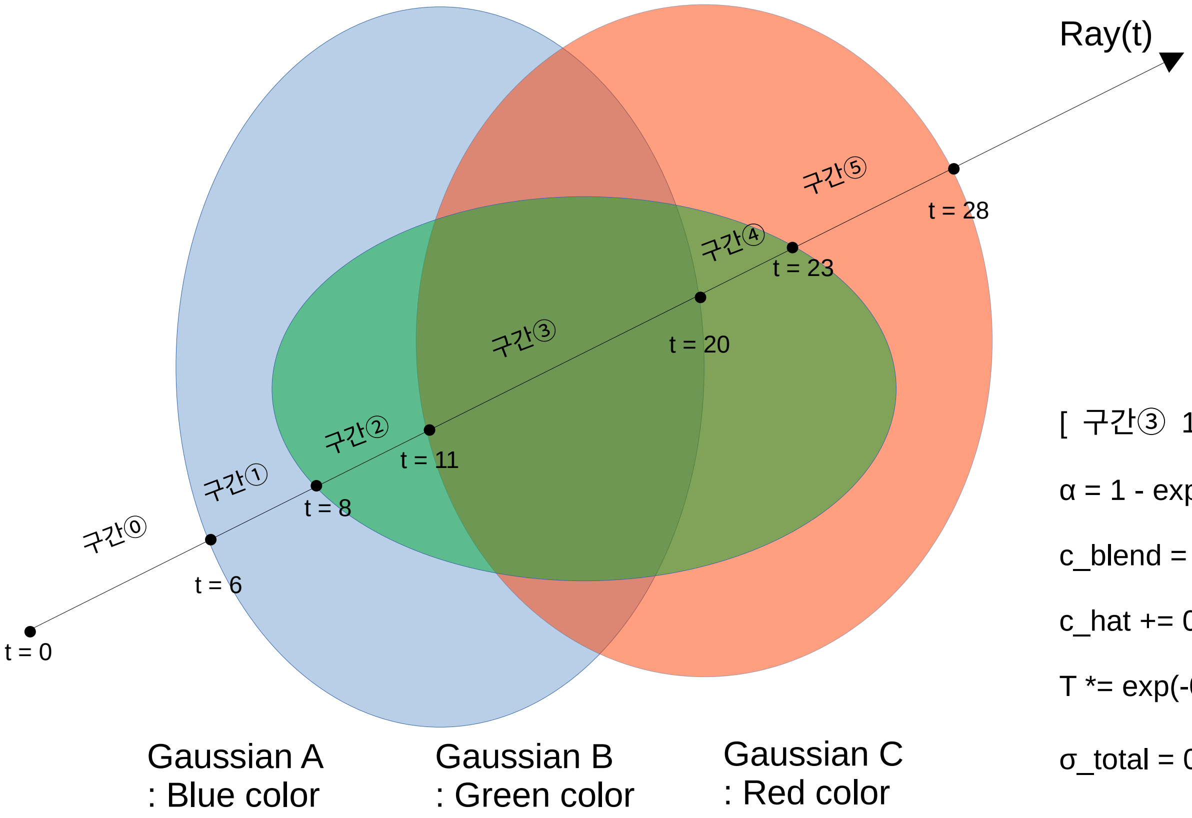
$c\_blend = (0.1 \times Blue + 0.1 \times Green) / 0.2 = (Blue + Green) / 2$

$c\_hat \mathrel{+}= 0.819 \times 0.451 \times (Blue+Green)/2$

$T \mathrel{*}= \exp(-0.2 \times 3) = 0.819 \times 0.549 \rightarrow 0.449$

σ_total = 0.2 --> +C 진입으로 0.2 + 0.1 = 0.3

EVER Overlap Gaussian



$\alpha = 1 - \exp(-\sigma \cdot \Delta t)$: 구간 (t) 알파

$c\_blend = (\sigma\_A \cdot c\_A + \sigma\_B \cdot c\_B + \ldots) / \sigma\_total$: 구간 색

$c\_hat \mathrel{+}= T \cdot \alpha \cdot c\_blend$: ray 전구간 누적 색

Event list : {A=6, B=8, C=11, -A=20, -B=23, -C=28}

각 가우시안의 σ(밀도) 는 0. 1 로 가정

[구간③ 11 → 20] active={A, B, C} $\Delta t = 20-11 = 9$

$\alpha = 1 - \exp(-0.3 \times 9) = 1 - 0.067 = 0.933$

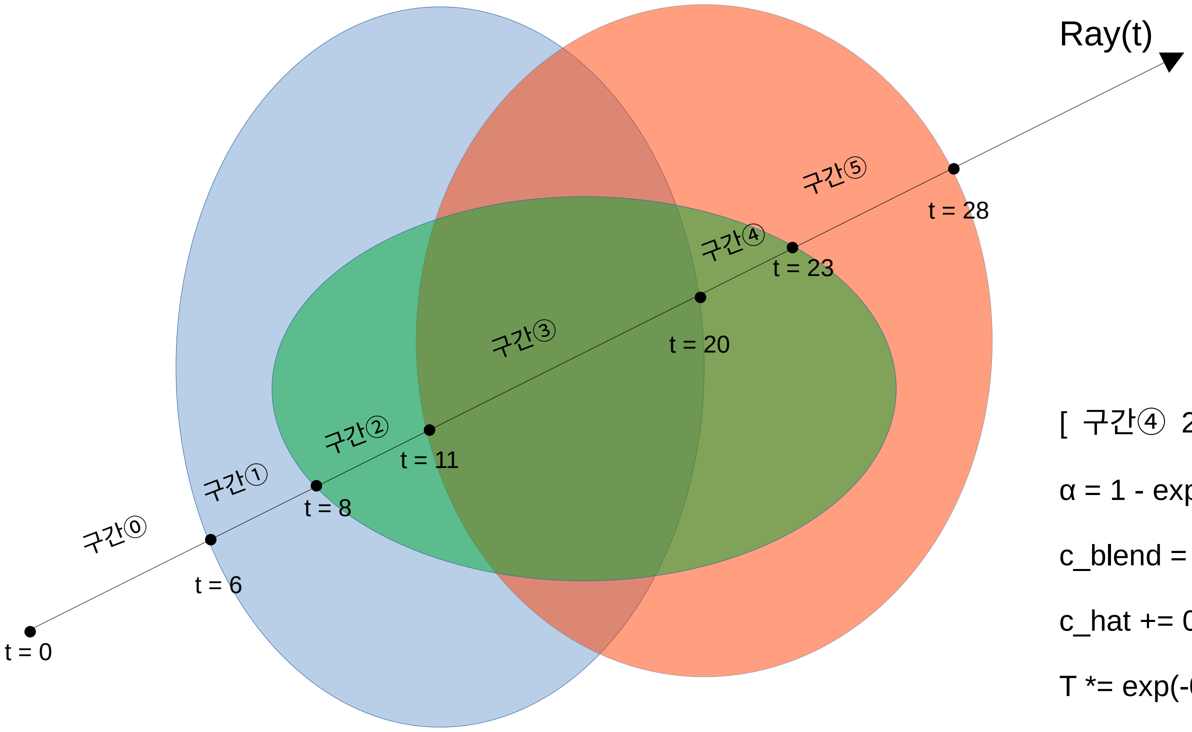
$c\_blend = (0.1\times Blue + 0.1\times Green + 0.1\times Red) / 0.3 = (Blue + Green + Red) / 3$

$c\_hat \mathrel{+}= 0.449 \times 0.933 \times (Blue+Green+Red)/3$

$T \mathrel{*}= \exp(-0.3 \times 9) = 0.449 \times 0.067 \rightarrow 0.030$

σ_total = 0.3 --> -A 탈출로 0.3 - 0.1 = 0.2

EVER Overlap Gaussian



Gaussian A
: Blue color

Gaussian B
: Green color

Gaussian C
: Red color

$\alpha = 1 - \exp(-\sigma \cdot \Delta t)$: 구간 (t) 알파

$c\_blend = (\sigma\_A \cdot c\_A + \sigma\_B \cdot c\_B + \ldots) / \sigma\_total$: 구간 색

$c\_hat \mathrel{+}= T \cdot \alpha \cdot c\_blend$: ray 전구간 누적 색

Event list : {A=6, B=8, C=11, -A=20, -B=23, -C=28}

각 가우시안의 σ(밀도) 는 0. 1 로 가정

[구간④ 20 → 23] active={B, C} $\Delta t = 23-20 = 3$

$\alpha = 1 - \exp(-0.2 \times 3) = 1 - 0.549 = 0.451$

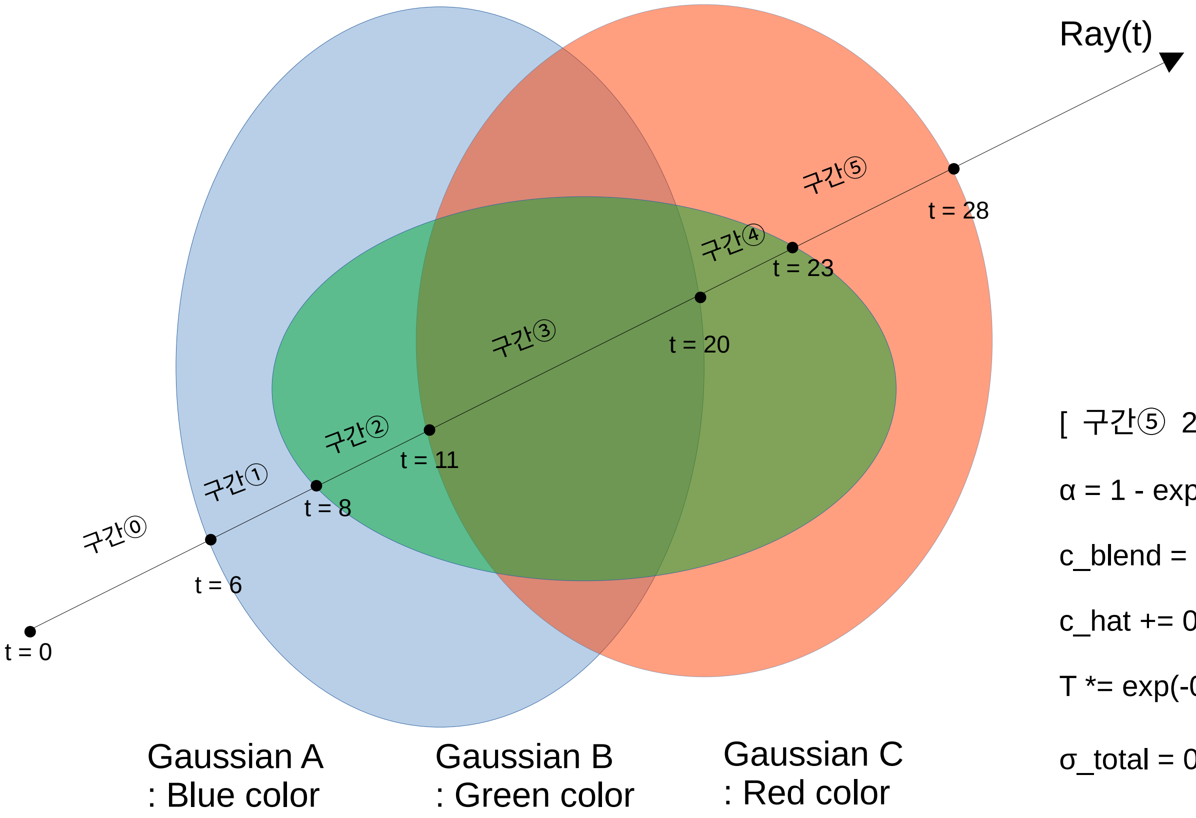
$c\_blend = (0.1 \times Green + 0.1 \times Red) / 0.2 = (Green + Red) / 2$

$c\_hat \mathrel{+}= 0.030 \times 0.451 \times (Green+Red)/2$

$T \mathrel{*}= \exp(-0.2 \times 3) = 0.030 \times 0.549 \rightarrow 0.016$

$\sigma\_total = 0.2 \rightarrow$ -B 탈출로 $0.2 - 0.1 = 0.1$

EVER Overlap Gaussian



α = 1 - exp(-σ · Δt) : 구간 (t) 알파

c_blend = (σ_A·c_A + σ_B·c_B + ...) / σ_total : 구간 색

c_hat += T · α · c_blend : ray 전구간 누적 색

Event list : {A=6, B=8, C=11, -A=20, -B=23, -C=28}

각 가우시안의 σ(밀도) 는 0. 1 로 가정

[구간⑤ 23 → 28] active={C} Δt = 28-23 = 5

α = 1 - exp(-0.1 × 5) = 1 - 0.607 = 0.393

c_blend = Red

c_hat += 0.016 × 0.393 × Red

T *= exp(-0.1 × 5) = 0.016 × 0.607 → 0.010

σ_total = 0.1 → -C 탈출로 0 (종료)

C_pixel = c_hat